

Pratik Singh

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Summary

Third-year B.Tech student at **NIT Jalandhar** focused on AI infrastructure, developer tools, and retrieval systems. Building repository intelligence platforms and context optimization infrastructure that improve how developers and AI coding agents reason over large software systems.

Education

National Institute of Technology Jalandhar

B.Tech – Electronics and Communication Engineering

– Coursework: Data Structures & Algorithms, Machine Learning, Web Application Development, Signal Processing

Jalandhar, India

Aug 2023 – May 2027

Experience

Flutter Developer Intern

VAYUXI ERP Pvt. Ltd.

- Architected reusable ERP modules supporting multiple business workflows through modular Flutter component architecture, reducing redundant UI code across role-based feature sets.
- Integrated RESTful APIs and Firebase services for real-time enterprise operations; optimized rendering pipelines and state management to improve application responsiveness under load.
- Maintained production codebase across Android and iOS via structured Git workflows, code reviews, and Agile delivery cycles.

3rd Place – OnDemand X Hackathon

NIT Jalandhar

- Led a developer team to design and integrate a real-time AI personalization model into OnDemand OS, delivering a working AI-centric OS prototype under competition constraints.

Projects

Repository Intelligence Platform (RIP)

Python · Tree-sitter · Neo4j · Qdrant · ONNX Runtime · FastAPI · Docker

- Designed a repository intelligence platform that transforms large multi-language codebases (Python, TypeScript, Dart, Java, Go, Rust) into a queryable architecture graph, enabling AI coding agents to reason over software structure rather than raw source files.
- Architected a hybrid retrieval engine combining Neo4j graph traversal, Qdrant vector search, and cross-encoder reranking to surface architecture-aware context – dependency graphs, workflow maps, and impact chains – while minimizing LLM context consumption.
- Optimized the indexing pipeline with parallel AST parsing via Tree-sitter and batched ONNX Runtime inference, achieving **9x faster** embedding generation; scaled to **589 files, 2K+ entities, and 5K+ relationships**.
- Exposed the knowledge graph through an **MCP server**, enabling AI coding agents to retrieve scoped architectural context instead of entire repositories.

Context Gateway – LLM Context Optimization Layer

Python · FastAPI · MCP · PostgreSQL · Redis · LiteLLM · Docker

- Identified that AI coding agents waste significant portions of context windows through redundant MCP tool calls and repeated cross-session retrieval; designed an orchestration layer that consolidates multiple invocations into a single token-budgeted context request.
- Implemented an intent classification and multi-source parallel retrieval pipeline with relevance ranking and context compression, maximizing high-value information density within fixed LLM context windows.
- Engineered session memory, conflict detection, role-based context filtering, and distributed caching to prevent redundant retrieval across concurrent AI agent sessions.

Cross-Platform ERP Platform

Flutter · Dart · Firebase · REST APIs · Android · iOS

- Architected a scalable modular ERP application with reusable component libraries, role-based feature modules, and real-time Firebase synchronization supporting live enterprise operations.
- Engineered offline-tolerant state management and rendering optimizations to maintain responsive performance across device classes in production.

Technical Skills

Languages: Python, Dart, JavaScript, C++

AI Systems: ONNX Runtime, Sentence Transformers, LiteLLM, MCP, Retrieval Pipelines, Cross-Encoder Reranking

Storage: Neo4j, Qdrant, PostgreSQL, Redis, MongoDB, MySQL

Backend: FastAPI, Node.js, Express.js, REST APIs, SQLAlchemy, Tree-sitter

Developer Infrastructure: Docker, Git, Linux, Firebase, Android Studio, Postman, Jupyter

Achievements

- **150+ DSA problems** solved across LeetCode and GeeksforGeeks.
- Delivered a full-stack NGO donation platform with payment integration and reporting within a **72-hour** freelance deadline.